

Review article

Blockchain for data spaces: A survey of architectures, capabilities, and challenges

Jianfeng Shi^a, Lianggen Xu^b, Feiwen Li^b, Lingyuan Yin^a, Zihang Yin^a, Yang Liu^{a,*}

^a China Academy of Information and Communications Technology, Beijing, China

^b Guangxi Beitou IT Innovation Technology Investment Group Co., Ltd., Nanning, China

ARTICLE INFO

Keywords:

Data space

Blockchain

Decentralized data sharing

Data sovereignty

Data governance

ABSTRACT

Data spaces have emerged as a promising paradigm for cross-organizational data sharing; however, their practical deployment still faces multiple technical and governance challenges. This paper presents a comprehensive survey of how blockchain technologies can support the construction and operation of data spaces. Specifically, we first analyze the key challenges encountered in existing data space initiatives, including identity verification, data sovereignty, assetization, contract fulfillment, compliance, governance, security, and cross-space interoperability. We then synthesize seven fundamental capabilities of blockchain that are relevant to addressing these challenges. Based on this capability-challenge perspective, we further organize and conceptualize a layered architectural framework for blockchain-enabled data spaces, consisting of identity, asset, circulation, usage, application, security, governance, and cross-space layers, illustrating how blockchain mechanisms may support each layer. A comprehensive review of representative academic studies and industrial practices is conducted to examine the current design patterns and technological approaches in this area. Finally, the paper summarizes open research challenges and outlines future research directions for integrating blockchain technologies into data spaces. Overall, this survey provides a structured perspective and research roadmap for the design of trustworthy, interoperable, and sustainable data spaces.

1. Introduction

Data spaces have emerged as a promising paradigm for secure and cross-organizational data sharing. The concept originated in Europe: in 2014, the German Fraunhofer-Gesellschaft proposed the Industrial Data Space to address challenges in industrial data exchange. Unlike traditional centralized data-sharing methods, data spaces aim to establish a trusted environment that enables efficient data exchange while preserving data sovereignty. European initiatives such as Gaia-X,¹ the International Data Spaces Association (IDSA),² and Catena-X³ have significantly advanced the development of data spaces, positioning them as a key direction for modern data-sharing infrastructures.

The interest in data spaces is growing worldwide. In China, the national data administration has proposed establishing over 100 trusted data spaces by 2028.⁴ In Europe, common data spaces are being devel-

oped across 14 sectors, including finance, manufacturing, and mobility. According to the IDSA Data Space Radar,⁵ 235 data spaces have been identified globally. Data spaces have been applied in enterprise data management [1], transportation [2,3], healthcare [4,5], manufacturing [6,7], supply chains [8], energy systems [9], AI applications [10] and other scenarios.

Despite this progress, implementing data spaces faces significant barriers. Challenges remain in establishing trust among participants, automating contract execution, ensuring governance, maintaining security, and achieving interoperability across heterogeneous systems. Blockchain technology, with its decentralized, immutable, and programmable nature, offers a promising foundation for addressing these

⁵ <https://digital-strategy.ec.europa.eu/en/policies/data-spaces>.

* Corresponding author.

E-mail addresses: shijianfeng@caict.ac.cn (J. Shi), xulg@bgigc.com (L. Xu), lifw@bgigc.com (F. Li), yinlingyuan@caict.ac.cn (L. Yin), yinzihang@caict.ac.cn (Z. Yin), liuyang7@caict.ac.cn, liuyangcaict@foxmail.com (Y. Liu).

¹ <https://gaia-x.eu/>

² <https://internationaldataspaces.org/>

³ <https://catena-x.net/>

⁴ <https://www.nda.gov.cn>

<https://doi.org/10.1016/j.rineng.2026.111927>

Received 12 March 2026; Received in revised form 30 June 2026; Accepted 9 July 2026

Available online 11 July 2026

2590-1230/© 2026 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

challenges. Integrating blockchain into data spaces can enhance trust, enable automated and verifiable transactions, and promote interoperable and sustainable ecosystems for data sharing.

However, the integration of blockchain into data spaces is still at an early stage. Existing studies tend to address blockchain and data spaces from partial and isolated perspectives, rather than examining how blockchain capabilities can be systematically mapped to data-space architectures and operational requirements. For example, Dam et al. [11] surveyed open-source data-space connectors with a focus on architectural implementation and usage-control support, while Siska et al. [12] reviewed technology stacks such as EDC, GXFS, and Pontus-X. Atzori et al. [13] compared data spaces with other data management solutions, such as data lakes, data warehouses, and databases, but did not specifically analyze blockchain-enabled design choices. Other works have examined specific dimensions of data spaces, including requirements [14], governance [15], and regional implementation differences between Japan and Europe [16]. Meanwhile, blockchain-related studies in data spaces have largely remained fragmented, focusing on isolated functions such as identity management [17], data tokenization [18], or smart-contract-based usage enforcement [19], without providing an integrated architectural view.

To address this gap, this paper proposes a layered architecture that systematically links blockchain capabilities with the construction and operation of data spaces. By connecting existing challenges, blockchain functions, and architectural layers in a unified framework, this survey moves beyond fragmented discussions and offers a more structured basis for comparing existing studies and identifying their complementary roles and limitations in supporting multi-stakeholder data ecosystems.

The main contributions of this paper are threefold.

- **Identification of Core Challenges.** Eight key challenges are systematically identified in the data space, covering identity trust, data sovereignty, assetization, contract fulfillment, compliance, governance, security, and cross-space interoperability.
- **Proposal of a Layered Blockchain-Enabled Architecture.** An eight-layer framework is proposed for blockchain-enabled data spaces, encompassing identity, asset, circulation, usage, application, security, governance, and cross-space layers, highlighting how blockchain can strengthen each layer.
- **Architecture-driven analysis and validation.** Existing research and industrial practices are analyzed within this architecture, providing insights into current design patterns, adoption trends, and open research challenges.

The remainder of this paper is organized as follows. Section 2 identifies the key challenges of data spaces. Section 3 summarizes the blockchain capabilities relevant to data spaces. Section 4 presents the proposed layered architecture and reviews corresponding studies for each layer. Section 5 explores open challenges and future research directions. Section 6 concludes the paper.

2. Challenges facing data spaces

2.1. Architecture

A data space can be defined as a decentralized, open, and data-sovereignty-oriented infrastructure that enables controlled and secure data exchange. Its core principle is to allow data providers and consumers to share data while maintaining autonomy over their assets.

Several international initiatives have proposed reference architectures to support data-space deployment, including IDSA, Gaia-X,⁶ FIWARE,⁷ and DSBA.⁸ Fig. 1 illustrates a typical data-space architecture, where connectors serve as terminals for both providers and consumers.

Data providers register metadata for their data products in the platform's data catalog via connectors. Consumers can search for desired datasets, verify identities, negotiate data-use contracts (including policies, pricing, and permissions), and subsequently access the data according to contractual agreements. This architecture facilitates standardized data sharing while preserving traceability and governance.

Current open-source connector implementations include the DSC connector,⁹ EDC connector,¹⁰ TRUE connector,¹¹ and Trusted connector.¹² When joining a data space, connectors create digital identities, register metadata, negotiate contracts, and handle data transfer and usage tracking, ensuring compliance with agreed-upon policies.

2.2. Challenges

Based on the architecture described above, we summarize eight major challenges in building and operating data spaces.

Challenge 1: Establishing Trusted Identities

Participants in data spaces must reliably authenticate each other. Existing mechanisms often rely on centralized authorities for credential issuance and validation, creating single points of failure and raising privacy and security concerns. Ensuring identity authenticity while protecting privacy in a multi-party environment remains a fundamental challenge.

Challenge 2: Clarifying Data Rights

Data ownership, usage, and operational rights can become ambiguous after multiple exchanges or transfers, leading to potential misuse. Current verification methods depend heavily on centralized institutions, limiting transparency and trust. Achieving decentralized, verifiable, and transparent data-rights management is therefore critical.

Challenge 3: Enabling Data Assetization

Data increasingly function as economic assets, capable of being priced, traded, or pledged. However, challenges such as unclear valuation, opaque transactions, uncertain ownership, insufficient profit distribution mechanisms, and low liquidity hinder effective data circulation. A trusted and transparent framework for data asset management is essential to support valuation, traceability, and equitable distribution.

Challenge 4: Automating Contract Fulfillment

Data-sharing agreements typically involve contracts specifying fees, access rules, and usage restrictions. Traditional contract execution is often manual or centralized, resulting in inefficiency, opacity, high transaction costs, and credit risk. Implementing automated, transparent, and trustworthy contract fulfillment mechanisms is crucial to streamline operations and minimize disputes.

Challenge 5: Enhancing Compliance Audits

Centralized compliance audits are prone to tampering, delayed reporting, and limited real-time verification. The complexity and opacity of traditional audit processes reduce regulatory effectiveness. Ensuring transparent, auditable, and decentralized compliance monitoring is therefore a significant challenge.

Challenge 6: Decentralizing Governance Mechanisms

Traditional governance models rely on centralized authorities to set standards, compliance frameworks, and incentives, leading to opaque decision-making, limited stakeholder participation, and slow adaptation. Designing fair, transparent, and participatory governance mechanisms is essential to align stakeholder interests and ensure sustainable data-space operations.

Challenge 7: Strengthening Security and Resilience

Critical components of a data space—such as data catalogs, semantic hubs, and contract management systems—are vulnerable to single points

⁹ <https://github.com/International-Data-Spaces-Association/DataspaceConnector>

¹⁰ <https://github.com/eclipse-edc/Connector>

¹¹ <https://github.com/Engineering-Research-and-Development/true-connector>

¹² <https://github.com/Fraunhofer-AISEC/trusted-connector>

⁶ <https://gaia-x.eu/>

⁷ <https://github.com/FIWARE/data-space-connector>

⁸ <https://data-spaces-business-alliance.eu/>

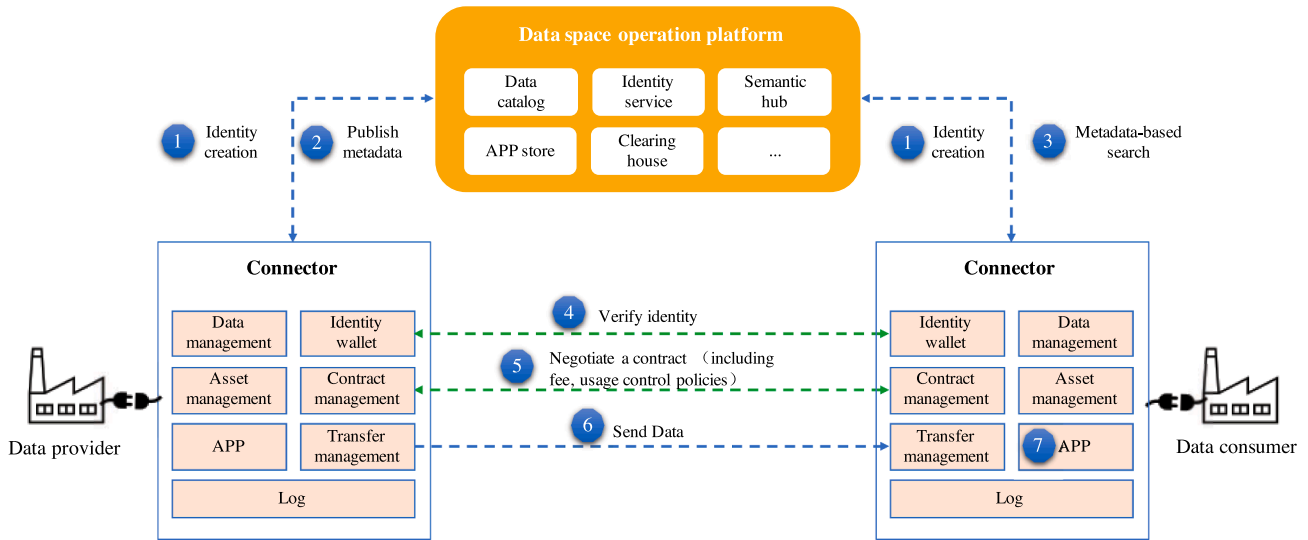


Fig. 1. Typical architecture of data spaces.

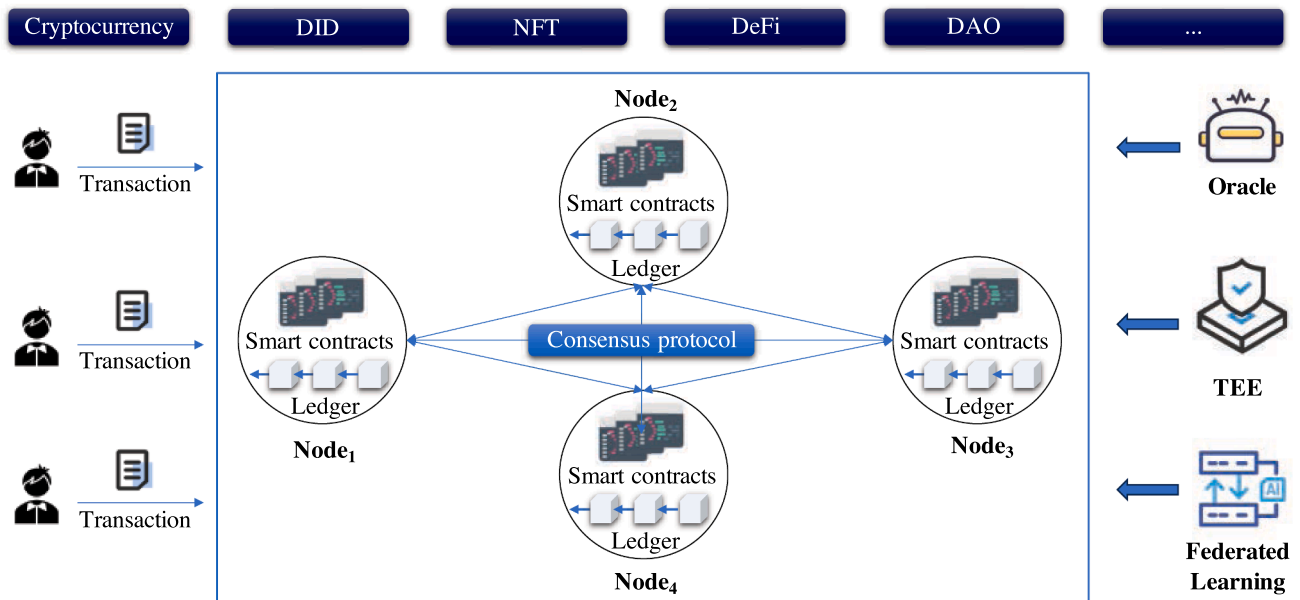


Fig. 2. Typical architecture of blockchain.

of failure and cyberattacks. Distributed denial-of-service (DDoS) attacks or tampering can disrupt services and undermine trust. Enhancing system security, fault tolerance, and resilience against malicious threats is a pressing requirement.

Challenge 8: Achieving Cross-Space Interoperability

Interoperability among data spaces remains limited. Protocols such as the Data Space Protocol (DSP)¹³ primarily address catalog exchange, contract negotiation, and data transfer. Heterogeneous standards and inconsistent policies hinder seamless cross-space collaboration. Building

mechanisms that ensure reliable and trustworthy interoperability across diverse data spaces is a major challenge.

3. Blockchain capabilities for data spaces

3.1. Architecture

Blockchain is a decentralized, distributed ledger technology that employs cryptographic algorithms to link data blocks chronologically, forming an immutable chain structure. Consensus protocols, such as Proof of Work (PoW) [20], Proof of Stake (PoS) [21], and Byzantine Fault Tolerance (BFT) [22], enable participating nodes to collectively validate transactions and maintain ledger consistency.

¹³ <https://docs.internationaldataspaces.org/ids-knowledgebase/dataspace-protocol>

As illustrated in Fig. 2, a blockchain network comprises multiple nodes, each maintaining a full copy of the ledger. Any modification to data must be validated by a majority of nodes, ensuring immutability and collective trust. This decentralized trust model allows value exchange and reliable collaboration among participants without centralized intermediaries.

Representative blockchain platforms include public chains such as Bitcoin [23] and Ethereum [24], as well as consortium blockchains such as Hyperledger Fabric [25], FISCO BCOS [26], HyperChain [27], and ChainMaker.¹⁴

3.2. Capabilities

Owing to its decentralized and programmable nature, blockchain introduces novel capabilities that can address many challenges in data spaces.

Capability 1: Identity Trust

Blockchain enables the establishment of independent, verifiable digital identities using asymmetric cryptography. Each participant holds a public-private key pair: the private key signs transactions, while the public key allows others to verify identity and transaction authenticity. This decentralized verification eliminates reliance on third-party authorities, reducing the risk of identity fraud and unauthorized access. Because identity data are replicated across multiple nodes, tampering requires simultaneous modification of all copies, ensuring data integrity and non-repudiation.

Capability 2: Trusted Data Recording

Blockchain ensures that on-chain data remain tamper-resistant, traceable, and auditable. Each block contains the hash of its predecessor, forming a continuous, immutable chain. Any modification alters subsequent hash values, and successful tampering would require control of the majority of network computing power, making attacks practically infeasible. This structure provides full traceability of data operations, from creation to current state, enabling reliable auditing and accountability.

Capability 3: Efficient Value Circulation

Blockchain supports tokenization and digitization of real-world assets (RWA) [28]. Tokens represent rights or assets such as stocks, bonds, or tickets-secured on-chain. RWA extends this mechanism by converting physical assets into tradable digital forms, allowing fractional ownership and rapid exchange. This lowers investment thresholds, enhances liquidity, and facilitates seamless circulation of value within digital ecosystems.

Capability 4: Rule Auto-Enforcement

Smart contracts allow business rules to be codified and executed automatically on the blockchain. Once predefined conditions are met, contracts execute without human intervention, ensuring consistent, reliable, and timely rule enforcement. For example, in supply chain management, smart contracts can automatically trigger payments or shipments based on quality verification and schedule compliance, improving coordination efficiency across multiple stakeholders.

Capability 5: Multi-Party Collaboration

Blockchain provides a transparent, auditable environment for complex, multi-stakeholder collaboration. Participants can define on-chain governance rules—including rights, obligations, and operational procedures through smart contracts, which automate execution and enforce agreements. Voting and incentive mechanisms further enhance fairness and accountability, fostering credible and traceable cooperation. This capability also supports emerging decentralized digital ecosystems that require collaborative governance across multiple stakeholders. For example, Alsamhi et al. [29] proposed a blockchain-driven decentralized architecture for metaverse data sharing, where smart contracts, decentralized autonomous organizations (DAOs), and blockchain-based gov-

Blockchain Capabilities	P1: Identity Trust	P2: Trusted Data Recording	P3: Efficient Value Circulation	P4: Rule Auto-Enforcement	P5: Multi-Party Collaboration	P6: Privacy Protection	P7: Security Defense
C1: Establishing Trusted Identities	•						
C2: Clarifying Data Rights		•					
C3: Enabling Data Assetization			•				
C4: Automating Contract Fulfillment				•			
C5: Enhancing Compliance Audits		•				•	
C6: Decentralizing Governance Mechanisms				•	•		
C7: Strengthening Security and Resilience						•	•
C8: Achieving Cross-Space Interoperability	•	•		•			

Fig. 3. Mapping between data space challenges and blockchain capabilities.

ernance mechanisms enable transparent, user-centric collaboration and decentralized decision-making.

Capability 6: Privacy Protection

Blockchain can integrate with privacy-enhancing technologies such as zero-knowledge proofs (ZKPs) [30], secure multi-party computation (MPC) [31], homomorphic encryption (HE) [32], and trusted execution environments (TEEs) [33]. These technologies enable computation or verification without disclosing underlying data. ZKP proves possession of information without revealing it. MPC allows multiple untrusted parties to jointly compute results securely. HE supports computation directly on encrypted data, and TEE provides isolated hardware environments for secure data processing. Beyond cryptographic techniques, blockchain has also been combined with federated learning (FL) to enable privacy-preserving collaborative intelligence over decentralized data spaces. For example, Alsamhi et al. [34] proposed a blockchain-enabled FL framework for decentralized healthcare data sharing, where blockchain provides trust, integrity, and audibility, while FL enables collaborative model training without exposing raw patient data.

Capability 7: Security Defense

Blockchain offers a multi-layered security framework resistant to tampering, DDoS [35], Sybil attacks, and other threats. Distributed data replication prevents single-node failures from causing data loss or service disruption. Consensus protocols ensure that nodes behave correctly, while digital signatures and multi-signature schemes further enhance transaction and data security, reinforcing trust across participants.

3.3. Mapping between data space challenges and blockchain capabilities

Although previous sections separately discuss the challenges of data spaces and the capabilities of blockchain, understanding how these capabilities collectively address different challenges is essential for designing blockchain-enabled data spaces. As shown in Fig. 3, the proposed capability–challenge mapping reveals the many-to-many relationships between the two perspectives. This mapping establishes a unified conceptual foundation for the layered architecture proposed in Section 4 and provides a systematic perspective for analyzing blockchain applications in data spaces.

4. Layered architecture of blockchain-enabled data spaces

Building upon the blockchain capabilities discussed in Section 3.2, this section proposes a layered blockchain application architecture specifically designed for data spaces. As illustrated in Fig. 4, the architecture comprises eight interrelated layers: identity, asset, circulation, usage, application, security, governance, and cross-space layers, each representing a functional domain within the data-space ecosystem.

¹⁴ <https://docs.chainmaker.org.cn/v2.3.6/html/>

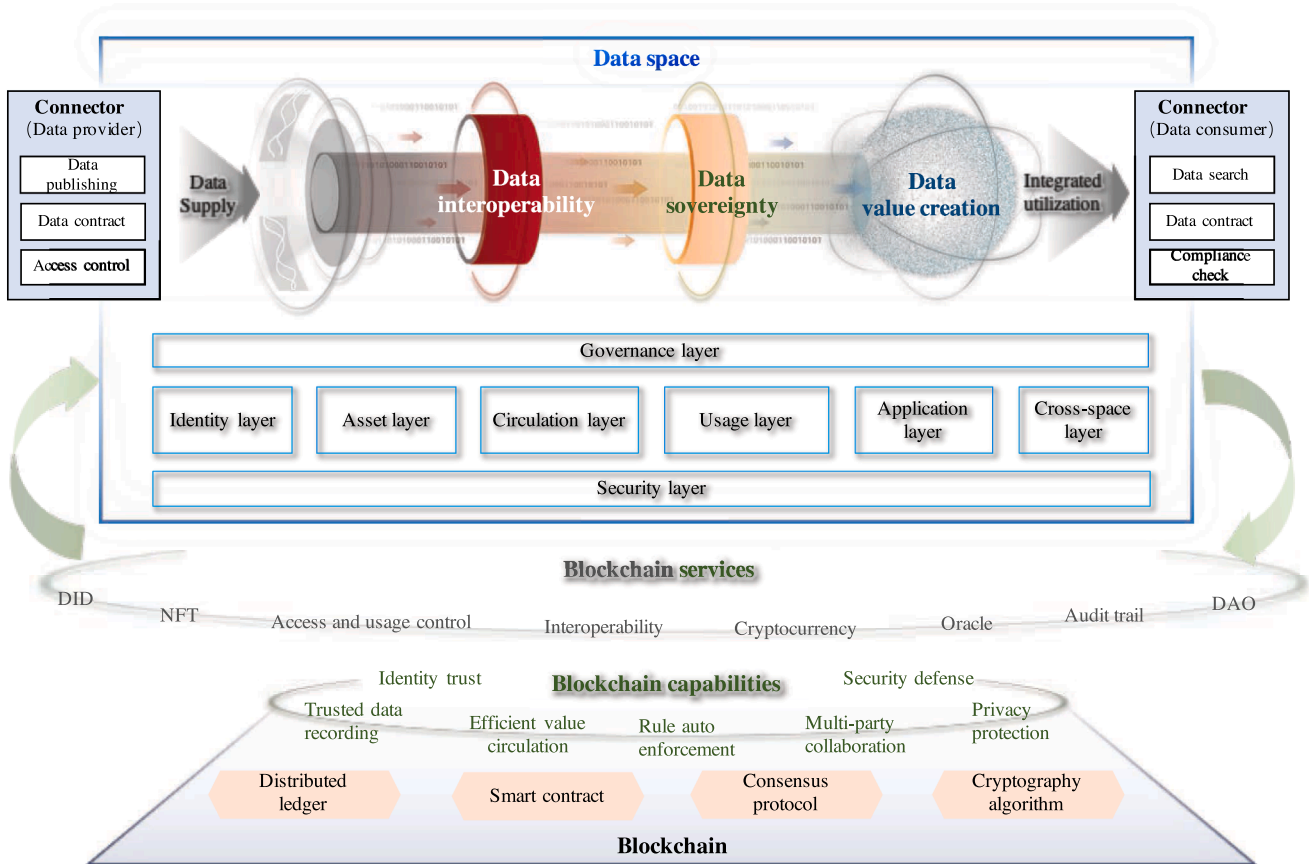


Fig. 4. Layered architecture of blockchain-enabled data spaces.

Existing studies corresponding to these layers are summarized in Table 1. The following subsections detail the role of blockchain in each layer and discuss representative research and industrial implementations.

4.1. Identity layer

The identity layer manages the identification, authentication, and authorization of all entities within the data space, including users (e.g., data providers and consumers), devices (e.g., connectors), and data products. Traditional identity management relies on centralized authorities (e.g., CAs or OAuth), which introduce single points of failure, privacy leakage risks, limited interoperability, and potential identity tampering or forgery.

Blockchain enables decentralized identity management by leveraging Decentralized Identifiers (DIDs) [79], Verifiable Credentials (VCs) [80], and Verifiable Presentations (VPs). This approach eliminates intermediaries, enhances privacy and security, and improves operational efficiency significantly.

As shown in Fig. 5, a DID is a self-sovereign digital identifier created and managed over a decentralized network, enabling entities to control and verify their identities autonomously. Each DID resolves to a DID document containing public keys, metadata, and service endpoints. During authentication, verifiers retrieve the DID document directly from the blockchain, ensuring authenticity and immutability without reliance on third-party certificate authorities.

A VC is a cryptographically signed digital certificate issued by a trusted authority (e.g., government, financial institution, or university) to attest to specific attributes of an entity, such as identity, qualifications, or creditworthiness. Holders retain full control of VCs locally,

while verifiers confirm their authenticity and integrity via blockchain records and issuer signatures, without direct interaction with the issuer. This mechanism enhances verification efficiency while preserving privacy.

A VP enables holders to present one or more VCs to a verifier, supporting selective disclosure and zero-knowledge proofs for privacy-preserving authentication. For example, a VP can allow an individual to prove they are over 18 without revealing the exact birth date, facilitating privacy-compliant verification across diverse scenarios.

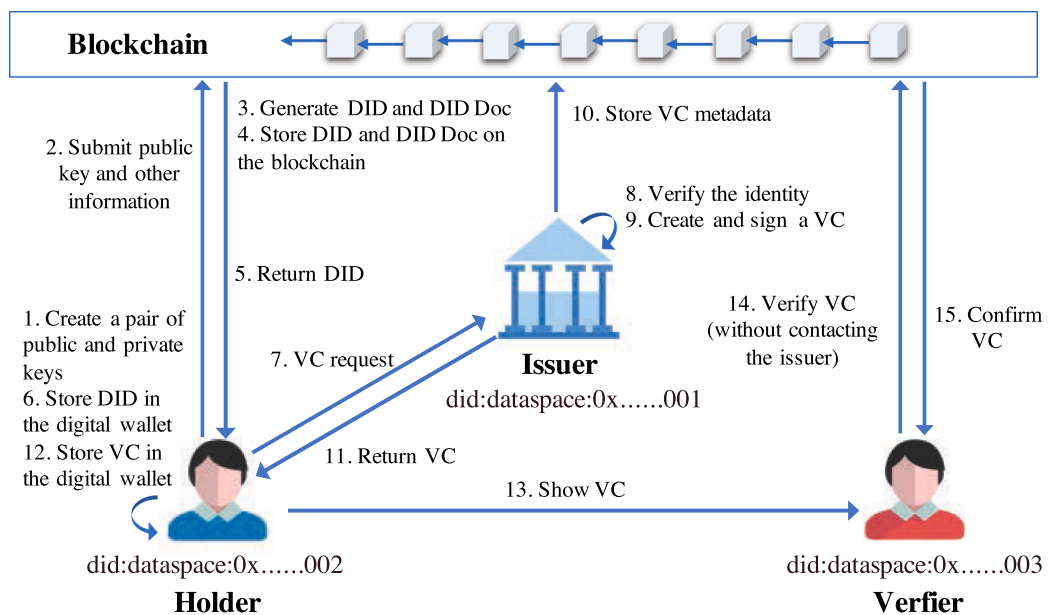
Existing studies on blockchain-based identity management in data spaces mainly fall into three categories: decentralized identity establishment, privacy-preserving credential management, and trust infrastructure construction. The first category focuses on decentralized identity establishment based on DIDs. For example, Klug et al. [36] introduced blockchain-based identifiers for products and stakeholders, while Karl et al. [38] deployed Ethereum-based DIDs to enhance participant authentication and dataset integrity. The second category emphasizes privacy-preserving identity management. García et al. [37] proposed an SSI framework enabling users to control personal data through decentralized registries, and Leander [39] compared different DID methods for SSI wallets in data-space environments. The third category focuses on blockchain-supported trust infrastructures. KRAKEN [40] and DSBA [41] validated decentralized identity verification through blockchain-based trust registries. Overall, existing studies have established a solid foundation for decentralized identity authentication and credential management, whereas comparatively less attention has been paid to cross-space identity interoperability and scalable trust federation.

Representative Use Case. The Blockchain Lab:UM developed a Decentralized Participant Discovery extension for the DSX Engine, where

Table 1

Existing work on the applications of blockchain in data spaces.

	Identity layer	Asset layer	Circulation layer	Usage layer	Application layer	Security layer	Governance layer	Cross-space layer
Klug et al. [36]	✓							
García et al. [37]	✓							
Karl et al. [38]	✓		✓					
Leander [39]	✓							
KRAKEN [40]	✓	✓		✓				
DSBA et al. [41]	✓					✓		
Yan et al. [42]		✓						
AMable [43]		✓						
Ocean Protocol [44]		✓		✓			✓	
Meneguzzo et al. [45,46]		✓						✓
Meneguzzo et al. [47]		✓						
Talmoudi et al. [48]			✓		✓			
Gallay et al. [49]			✓					
Diallo et al. [50]			✓					
Kyriakou et al. [51,52]				✓				
Alsamhi et al. [53,54]				✓				
Dutkiewicz et al. [55]				✓				
Sang et al. [56]					✓			
Gall et al. [57]					✓			
Liu et al. [58]					✓			
Bonomi et al. [59]						✓		
Kovach et al. [60,61]						✓		
Witharanage et al. [62]						✓		
Singh et al. [63]						✓		
Alsamhi et al. [64]						✓		
EDC [65]						✓		
Pontus-X [66]						✓		
IDS-RAM 4 [67]						✓		
Dalmolen et al. [68]						✓	✓	
Baraku et al. [69]						✓	✓	
Catena-x [70]								✓
VESPACE [71]	✓		✓					
Seike et al. [72]			✓					
Gentili [73]							✓	
Liang [74]							✓	
Villaverde [75]							✓	
Cano-Benito [76]					✓			
Yang [77]					✓			
Shi et al. [78]							✓	

**Fig. 5.** Blockchain applications in the identity layer of data spaces.

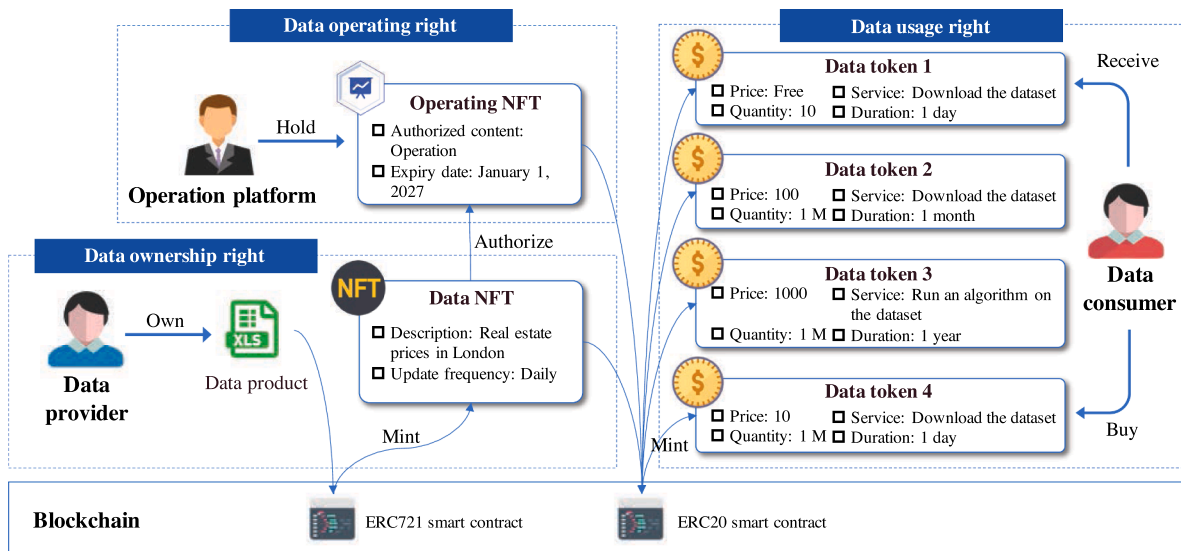


Fig. 6. Blockchain applications in the asset layer of data spaces.

each connector automatically registers and deregisters its DID through Ethereum smart contracts. Instead of relying on a centralized participant registry, connectors discover each other by monitoring blockchain events, demonstrating how blockchain can provide decentralized identity registration and participant discovery for trusted data-space membership.¹⁵

4.2. Asset layer

The asset layer manages the creation, verification, transfer, revocation, and trading of data-related rights—including ownership, operational rights, and usage permissions—to support the assetization and circulation of data assets. In traditional systems, data asset management typically relies on centralized platforms, leading to challenges such as ambiguous ownership, inefficient rights confirmation, restricted circulation, inflexible authorization mechanisms, and unfair benefit distribution.

Blockchain enables transparent and programmable data asset management by leveraging smart contracts and tokenization mechanisms. Through tokenized representations, blockchain allows data assets and their associated rights to be managed, traded, and audited in a decentralized environment. In particular, non-fungible tokens (NFTs) [81] can represent the unique ownership of data assets, enabling fine-grained authorization and programmable revenue-sharing models. Meanwhile, fungible data tokens can represent access, computation, or download rights, allowing data assets to be traded as digital commodities with verifiable ownership and clearly defined usage terms.

As illustrated in Fig. 6, the ERC-721 standard defines non-fungible tokens that are unique, indivisible, and transferable, making them suitable for representing individual data assets and their ownership records. Each NFT can embed the owner's DID together with metadata describing the dataset, such as its content, update frequency, and source (e.g., daily updates of London real estate prices). This design ensures transparent ownership and well-defined data characteristics. Data owners may further issue operational NFTs to authorized entities, specifying the scope of authorization (e.g., operational privileges) and the validity period (e.g., until January 1, 2027). Together, these NFTs support verifiable rights confirmation, traceable operations, and market-oriented data asset management.

In parallel, ERC-20-based data tokens can represent rights to access or utilize data services. Unlike NFTs, ERC-20 tokens are fungible and

highly liquid, enabling automated settlement and efficient management of data transactions. Each token can encode parameters such as price, quantity, service type (e.g., dataset download), and duration, thereby enabling flexible and programmable access strategies. For instance, data providers may distribute trial tokens to increase dataset visibility and attract potential consumers.

Existing studies on blockchain-enabled data asset management in data spaces mainly focus on three aspects: ownership confirmation, tokenized asset representation, and token-based value circulation. The first line of research leverages blockchain to establish trusted ownership and intellectual property protection. Yan et al. [42] applied blockchain to clarify data ownership within integrated resource systems, while the AMable project [43] anchored design-file hashes on-chain to provide tamper-resistant copyright protection. The second line emphasizes data asset tokenization. Ocean Protocol represents data assets using blockchain-based tokens, where NFTs serve as unique identifiers for ownership tracking and access control. Building upon this framework, Meneguzzo et al. [45,46] adopted NFTs and data tokens to manage energy datasets, and further utilized ERC-721 NFTs for copyright representation together with ERC-20 tokens as access licenses [47]. The third line explores token-enabled value circulation. For example, the KRAKEN project [40] adopted Streamr's DATA token as a payment medium operating across the Ethereum and xDai blockchains. Overall, existing studies have established effective mechanisms for ownership confirmation and tokenized representation of data assets, whereas comparatively less attention has been devoted to dynamic rights management, asset valuation, and lifecycle management of data assets across heterogeneous data spaces.

Representative Use Case. In the Horizon Europe DS2 – DataSpace, DataShare 2.0 project, Athens Technology Center developed a blockchain-based Data Rights Management module that records data-access events, transfers, and usage policies on blockchain. The module enables traceable ownership management and verifiable usage rights, illustrating how blockchain supports trusted lifecycle management of data assets across decentralized data spaces.¹⁶

4.3. Circulation layer

The circulation layer governs the transaction and delivery processes between data providers and consumers, ensuring that all exchanges

¹⁵ <https://medium.com/@blockchainlabum/introducing-dsx-engine-truly-decentralized-dataspace-connector-e40be91afe8c>

¹⁶ <https://www.atc.gr/atc-develops-ai-and-blockchain-technologies-for-european-dataspace-through-the-ds2-project/>

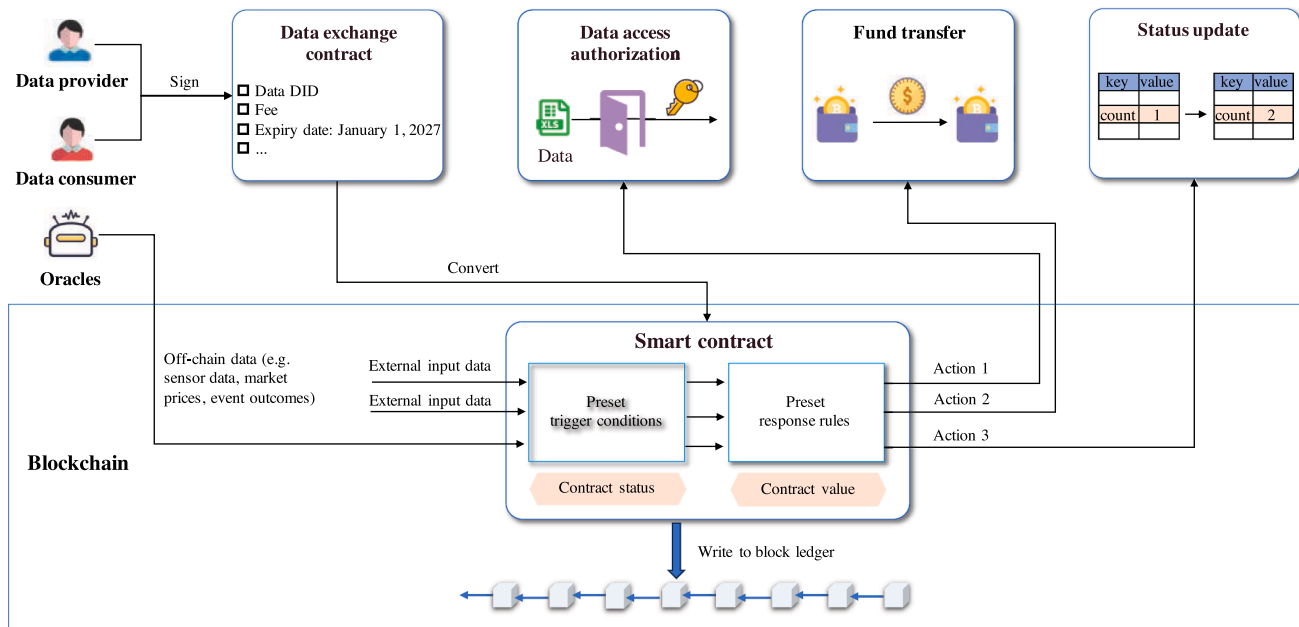


Fig. 7. Blockchain applications in the circulation layer of data spaces.

comply with predefined contractual agreements. In traditional systems, data circulation typically relies on centralized platforms or paper-based contracts. This approach results in high trust costs, opaque contract execution, limited traceability, reliance on intermediaries for settlement, and increased risks of contract breaches.

Blockchain addresses these limitations by transforming data-exchange agreements into self-executing smart contracts deployed on-chain. Through this mechanism, blockchain enables tamper-resistant contract management, automated execution, traceable data delivery, and transparent settlement. As a result, the security, efficiency, and compliance of data circulation can be significantly improved.

As illustrated in Fig. 7, data-exchange smart contracts are autonomous programs deployed on the blockchain that automatically manage access control, transaction states, and payment settlement. Each contract defines predefined trigger conditions and corresponding execution rules while maintaining its operational state and transaction records on-chain. Smart contracts can also interact with off-chain oracles [82], which provide external data inputs such as sensor readings, market prices, or event outcomes. These inputs enable automated operations including access authorization, transaction status updates (e.g., exchange counts), and fund transfers based on predefined triggers. All execution results are permanently recorded on the blockchain ledger, ensuring that the entire data-delivery process remains transparent, auditable, and verifiable. By eliminating intermediaries, smart contracts reduce operational costs while strengthening trust and efficiency in data circulation.

Existing studies on blockchain-enabled data circulation in data spaces mainly focus on three aspects: smart-contract-based transaction automation, verifiable access control, and performance optimization. The first line of research leverages smart contracts to automate data exchange and settlement. Talmoudi et al. [48] utilized smart contracts to automatically compensate data providers and streamline the acquisition of data usage rights. Similarly, Gallay et al. [49] and Diallo et al. [50] employed blockchain-based smart contracts to automate logistics transactions and payment settlement, while Karl et al. [38] applied contract-based automation to support transparent data exchange in public-security data spaces. The second line focuses on verifiable data circulation through trusted identity and access control. Costagliola et al. [71] integrated DIDs, VCs, and blockchain technologies to achieve verifiable data access control within data spaces. The third line investigates

the efficiency of blockchain-supported data circulation. Seike et al. [72] analyzed the impact of blockchain consensus mechanisms on system performance and proposed a batch queueing model with buffer capacity to evaluate the efficiency of blockchain-enabled data circulation. Overall, existing studies demonstrate the effectiveness of blockchain in automating trusted data circulation, while comparatively less attention has been paid to adaptive contract negotiation, cross-space transaction coordination, and scalable circulation mechanisms for large-scale data-space ecosystems.

Representative Use Case. China Telecom's Trusted Data Space integrates blockchain, privacy-preserving computation, and secure data sandboxes into a unified infrastructure, while blockchain provides immutable transaction records and auditable evidence throughout the data-sharing process.¹⁷

4.4. Usage layer

The usage layer governs how data consumers access and process data from multiple providers while ensuring privacy, security, and regulatory compliance. In traditional data-processing models, consumers typically download raw datasets for local computation. This approach increases the risks of data leakage, unauthorized usage, and unverifiable processing activities.

Blockchain enables verifiable and privacy-preserving data utilization through decentralized computation frameworks. By integrating privacy-enhancing technologies such as trusted execution environments (TEEs), secure multi-party computation (MPC), and federated learning (FL)-blockchain supports computation over distributed data without exposing the underlying datasets, thereby realizing the principle of "available but invisible" data usage. This paradigm protects sensitive information while maintaining transparency, accountability, and traceability throughout the computation lifecycle.

As illustrated in Fig. 8, data consumers first acquire data tokens from multiple providers and submit their computation models together with the corresponding tokens to each provider. Upon receiving the request, providers verify the consumer's access and usage rights, execute local computation or model training on their private datasets, and

¹⁷ <https://mp.weixin.qq.com/s/ZI6Y7Q7wMfOoicUUYSK9Mw>

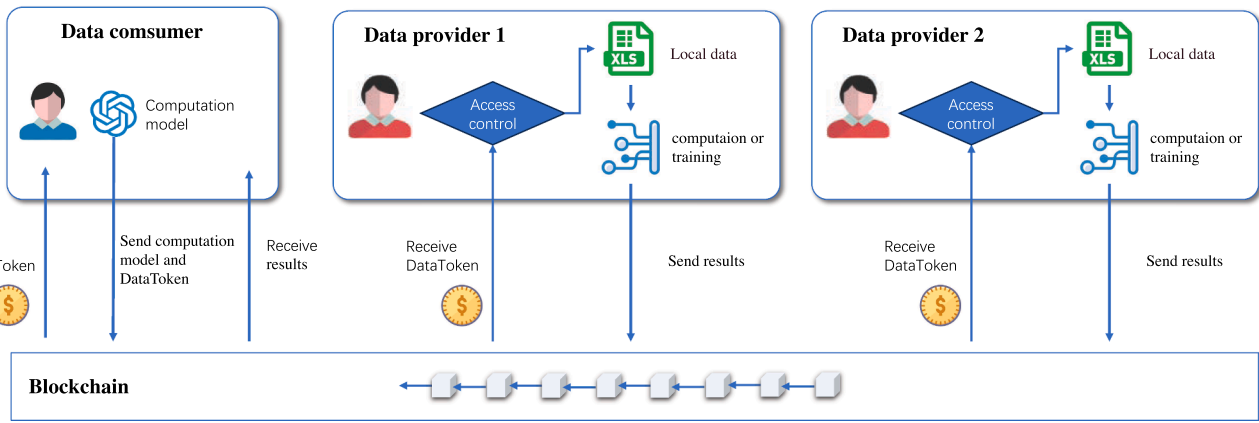


Fig. 8. Blockchain applications in the usage layer of data spaces.

return encrypted results to the consumer. The consumer then aggregates these partial outputs to generate the final result. All interactions—including authorization, computation execution, and result submission—are immutably recorded on the blockchain ledger, ensuring verifiable execution, compliant data usage, and auditable outcomes. This mechanism prevents raw data exposure, strengthens providers' control over their data assets, and enhances trust in cross-organizational collaborative computation.

Existing studies on blockchain-enabled data usage in data spaces mainly focus on three aspects: decentralized authorization, privacy-preserving computation, and usage policy enforcement. The first line of research leverages blockchain as a trusted authorization infrastructure for data access. Kyriakou et al. [51,52] developed a blockchain-based biometric data space in which smart contracts automate access control and prevent unauthorized revocation of data rights. Similarly, the KRAKEN project [40] employs blockchain as a decentralized authorization layer to approve qualified consumers for data access. The second line integrates blockchain with privacy-enhancing technologies to enable secure data computation. Alsamhi et al. [53,54] proposed a blockchain-enabled federated learning framework for drone data spaces, allowing collaborative model training without exposing raw data. Dutkiewicz et al. [55] further combined homomorphic encryption, MPC, and TEEs to strengthen privacy during smart-contract execution. The third line focuses on enforcing data usage policies through blockchain. Ocean Protocol integrates smart contracts to automatically enforce usage policies, enabling transparent and intermediary-free data transactions.¹⁸ Overall, existing studies have demonstrated the effectiveness of blockchain in enabling trusted authorization and privacy-preserving data usage, whereas adaptive usage control, fine-grained policy enforcement, and verifiable cross-space collaborative computation remain relatively underexplored.

Representative Use Case. The Anhui Provincial Trusted Data Space, developed by the Institute of Data Space of Hefei Comprehensive National Science Center, combines blockchain with Trusted Execution Environments (TEE). Blockchain provides end-to-end provenance and tamper-proof evidence for data circulation, while TEE ensures “usable but invisible” data processing, demonstrating a practical security architecture for cross-department data collaboration.¹⁹

4.5. Application layer

The application layer enables the development and orchestration of diverse data-driven services within blockchain-enabled data spaces. Tra-

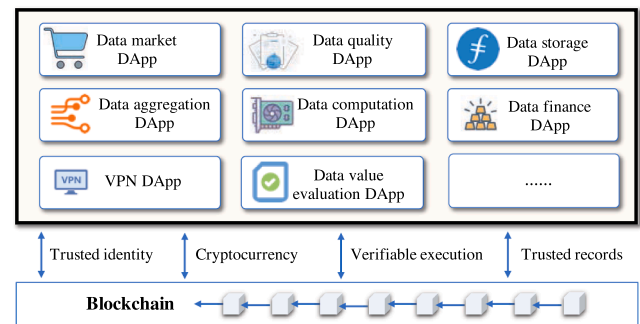


Fig. 9. Blockchain applications in the application layer of data spaces.

ditional data applications typically rely on centralized platforms or intermediaries for data integration, exchange, and service delivery. Such architectures often lead to opaque trust mechanisms, limited interoperability, security vulnerabilities, and single points of failure.

Blockchain addresses these limitations by supporting decentralized applications (DApps) that operate on distributed infrastructures and leverage smart contracts for automated service execution. These applications enable transparent coordination among multiple participants while ensuring verifiable operations and tamper-resistant service records.

As illustrated in Fig. 9, various DApps can be built on top of blockchain infrastructures to support trustworthy data services:

- Data marketplace DApps facilitate the decentralized registration, pricing, and trading of data assets through smart contracts, enabling transparent and auditable data exchanges.
- Data quality DApps perform integrity verification, trusted evidence recording, and quality assessment to ensure the reliability and traceability of shared datasets throughout their lifecycle.
- Data storage DApps utilize decentralized storage technologies such as IPFS [83], Arweave [84], and Filecoin to provide tamper-resistant, highly available, and persistent data storage across distributed nodes.
- Data aggregation DApps enable the integration and provenance tracking of datasets from heterogeneous sources, supporting the generation of new value-added data products.
- Data computation DApps encourage participants to contribute computing resources and reward them through data-token incentives, enabling scalable and cost-efficient computation across distributed data environments.
- Data finance DApps support the tokenization and financialization of data assets through mechanisms such as data collateralization, revenue sharing, and liquidity mining.

¹⁸ <https://docs.oceanprotocol.com/user-guides/basic-concepts#blockchain-the-backbone-of-ocean>

¹⁹ https://mp.weixin.qq.com/s/wA9YVGeHY2pO2uTM_4CwVA

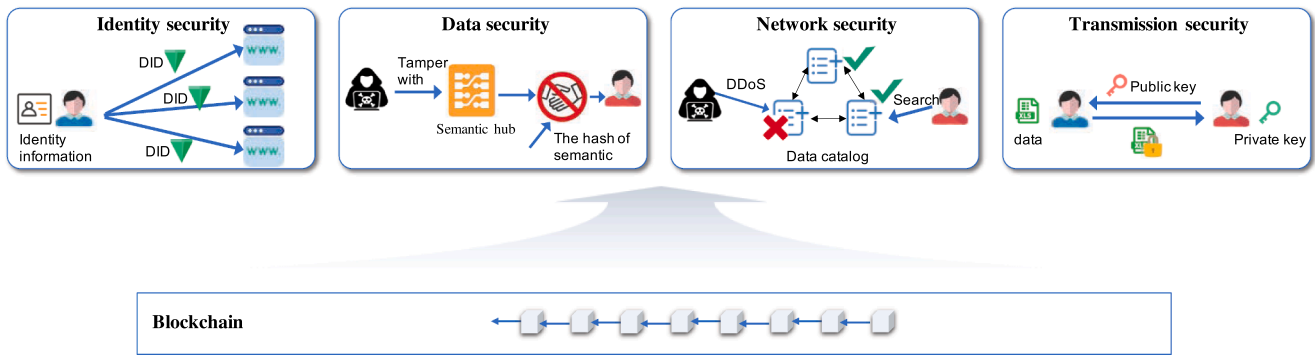


Fig. 10. Blockchain applications in the security layer of data spaces.

- VPN DApps incentivize bandwidth sharing via blockchain-based reward mechanisms, thereby forming a resilient, censorship-resistant, and incentive-driven data transmission infrastructure.

Existing studies have explored blockchain-enabled applications in data-space environments, mainly focusing on decentralized data infrastructure, trusted industrial applications, and secure data lifecycle management. The first line of research develops decentralized infrastructures to support trusted data services. Talmoudi et al. [48] integrated data spaces with IPFS to provide decentralized data containers for verifiable storage and retrieval within agent-based systems. The second line applies blockchain to domain-specific data-space applications. Sang et al. [56] developed a predictive-maintenance platform that combines blockchain infrastructures with IDS connectors to improve transparency, traceability, and auditability in collaborative manufacturing. Similarly, Gall et al. [57] proposed a blockchain-as-a-service audit logging system that enables IDS connectors to immutably record operational events and transaction metadata through standardized APIs. The third line focuses on secure management of the data lifecycle. Liu et al. [58] designed a trusted decentralized platform integrating blockchain and trusted execution environments (TEEs) to support secure data storage, pricing, querying, and privacy-preserving computation. Yang et al. [77] further proposed a blockchain-based secure storage framework to strengthen the trustworthiness and protection of data assets within data spaces. Overall, existing applications demonstrate the practicality of blockchain across diverse data-space scenarios, whereas most solutions remain domain-specific and provide limited support for reusable application frameworks, cross-domain interoperability, and large-scale deployment.

4.6. Security layer

The security layer safeguards data integrity, confidentiality, and availability throughout the entire lifecycle of data spaces. In traditional architectures, security management is typically centralized, which introduces vulnerabilities such as identity leakage, interception or tampering during data transmission, attacks on critical services (e.g., semantic hubs), and single points of failure—for instance, distributed denial-of-service (DDoS) attacks targeting centralized data catalogs.

Blockchain strengthens the security foundation of data spaces by introducing decentralized trust mechanisms, tamper-resistant records, and cryptographic verification. These capabilities enhance identity protection, data integrity assurance, network resilience, and secure data transmission across distributed environments.

Fig. 10 illustrates how blockchain-based mechanisms reinforce multiple security dimensions. Decentralized Identifiers (DIDs) and on-chain DID documents enable cross-platform authentication and non-repudiable identity verification, reducing dependence on centralized credential repositories. Semantic mappings and metadata can be anchored on-chain as cryptographic hashes, enabling tamper detection

and verifiable integrity validation. Additionally, distributing catalog metadata across blockchain nodes improves service availability and resilience against node-level attacks.

To protect communication confidentiality, data transmissions should employ end-to-end encryption, such as public-key cryptographic schemes, ensuring that only intended recipients can decrypt the payload. Message authentication mechanisms and on-chain hash anchoring further guarantee data integrity and provide verifiable evidence of data provenance.

Existing studies on blockchain-enabled security in data spaces mainly focus on three aspects: data integrity and traceability, secure infrastructure construction, and trust enhancement for distributed environments. The first line of research leverages blockchain to ensure data integrity and traceability. Bonomi et al. [59] developed an interoperable traceability platform that utilizes blockchain to guarantee data integrity across distributed systems. The second line strengthens the security of data-space infrastructures by integrating blockchain into trusted system components. Kovach et al. [60,61] implemented key IDS components, including identity providers, metadata catalogs, and clearing houses, on the DAG-based distributed ledger IOTA [85] to improve both security and usability. Similarly, Witharanage et al. [62] migrated trusted IDS entities to Hyperledger Fabric to enhance privacy protection and access control. The third line focuses on reinforcing trust in distributed environments. Singh et al. [63] proposed blockchain-enabled trust mechanisms for edge data spaces, enabling tamper-evident control messaging among edge peers. Alsamhi et al. [64] further emphasized the role of blockchain, together with public-private key cryptography, in strengthening the security of Data Space 4.0. Overall, existing studies demonstrate that blockchain effectively enhances the security and trustworthiness of data-space infrastructures, whereas comprehensive security architectures covering cross-space threat detection, dynamic trust assessment, and coordinated security governance remain relatively underexplored.

Representative Use Case. The Eclipse Dataspace Components (EDC)²⁰ framework introduces decentralized identity hubs to reduce single points of failure and maintain user control over credentials. Pontus-X²¹ anchors catalog metadata on-chain to guarantee integrity and reliability. Furthermore, IDS RAM^{4,22} and DSBA [41] discuss blockchain-based persistence layers—including EBSI-compatible chains—for recording shared catalogs and participant registries across national and cross-organizational data-space networks.

²⁰ <https://eclipse-edc.github.io/documentation/for-adapters/identity-hub/>

²¹ https://docs.pontus-x.eu/docs/ddo_credential/on_chain#on-chain-metadata---anchoring-of-oe-service-credentials.

²² https://docs.internationaldataspaces.org/ids-knowledgebase/ids-ram-4/context-of-the-international-data-spaces/2_1_data-driven-business-ecosystems/2_9_blockchain

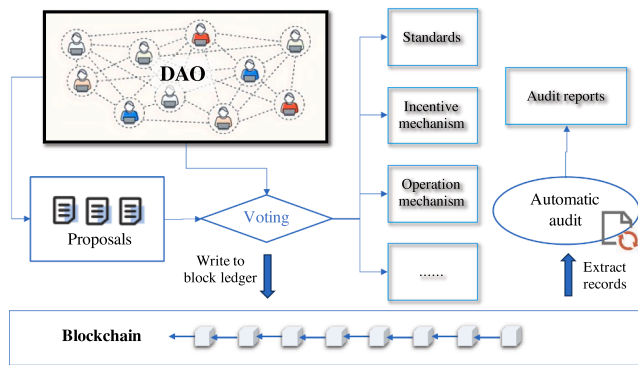


Fig. 11. Blockchain applications in the governance layer of data spaces.

4.7. Governance layer

The governance layer defines and enforces the rules, processes, and responsibilities among stakeholders in a data-space ecosystem to ensure compliant, fair, and accountable data sharing. In traditional systems, governance mechanisms are typically centralized, which often results in opaque decision-making, concentration of power, limited stakeholder participation, delayed or retrospective audits, and difficulties in reconciling conflicting interests among participants.

Blockchain introduces decentralized governance mechanisms that improve transparency, traceability, and collective participation in decision-making processes. By leveraging on-chain governance primitives—such as decentralized autonomous organizations (DAOs) [86], on-chain proposals, and verifiable voting—blockchain enables stakeholders to collaboratively define and enforce governance rules in a transparent and auditable manner.

As illustrated in Fig. 11, stakeholders can form DAOs to manage governance processes within data spaces. Members may submit proposals related to standards, incentive mechanisms, operational policies, or data-sharing rules. Smart contracts automatically manage proposal submission, voting procedures, and execution of approved decisions, significantly reducing manual intervention. All governance activities—including proposal creation, voting outcomes, and rule updates—are immutably recorded on the blockchain, providing a transparent and auditable governance history.

Existing studies on blockchain-enabled governance in data spaces mainly focus on three aspects: transparent governance and accountability, privacy-aware governance, and decentralized collaborative governance. The first line of research leverages blockchain to improve governance transparency and accountability. Dalmolen et al. [68] implemented a blockchain-backed registry for broker service providers to enhance transparency and auditability in service governance. Similarly, Baraku et al. [69] proposed a responsible data-control framework that combines blockchain-based transaction logs with provenance tracking to strengthen accountability, while Ocean Protocol records tamper-evident audit trails on-chain to enable verifiable governance of data transactions.²³ The second line addresses the tension between governance transparency and privacy protection. Since fully public traceability may conflict with the confidentiality requirements of data spaces, governance mechanisms increasingly incorporate privacy-preserving techniques such as selective disclosure and confidential voting to balance auditability with data sovereignty. The third line explores decentralized governance mechanisms. Shi et al. [78] proposed a DAO-based reputation mechanism that enables stakeholders to collaboratively update reputation scores through tamper-evident records and in-

centive alignment. Gentili [73] further discussed how blockchain can complement governance frameworks by enhancing transparency, accountability, and stakeholder participation. Overall, existing studies have demonstrated the value of blockchain in improving governance transparency and stakeholder collaboration, whereas practical governance frameworks that balance decentralization, privacy protection, regulatory compliance, and cross-space coordination remain relatively immature.

Representative Use Case. The DSX Engine implements a Decentralized Clearing House extension, where connectors continuously submit data-sharing events to Ethereum smart contracts. The blockchain maintains an immutable event log for recording data access, sharing, and usage activities, providing transparent auditing and accountability without relying on centralized logging services.²⁴

4.8. Cross-space layer

The cross-space layer provides the fundamental mechanisms required for trustworthy data exchange and service interoperability across heterogeneous data spaces. In traditional architectures, differences in identity systems, settlement infrastructures, and access-control policies often hinder interoperability, leading to fragmented authentication processes, cumbersome cross-space settlement, and inconsistent enforcement of data-usage constraints.

Blockchain technologies introduce decentralized trust infrastructures that enable secure and verifiable interaction across independent data spaces. By leveraging distributed identity management, programmable value exchange, and tamper-evident recordkeeping, blockchain can serve as an interoperability backbone for federated data-space ecosystems.

As illustrated in Fig. 12, blockchain facilitates trusted cross-space interaction through several key mechanisms. Decentralized Identifiers (DIDs) allow participants to authenticate across different data spaces without repeated registration, enabling cross-domain identity verification with cryptographic guarantees. Smart contract-based swap protocols enable automated value exchange between data spaces, reducing reliance on manual settlement processes and minimizing trust dependencies among heterogeneous blockchain environments.

To ensure data authenticity during cross-space transfer, consumers can verify data integrity by comparing cryptographic hashes against on-chain registration records, thereby providing tamper-evident validation of shared datasets. In addition, policy translation mechanisms enable usage-control rules to remain enforceable when data flows across domains with different governance policies. Operational events associated with cross-space interactions can also be recorded on-chain, creating transparent and auditable logs that support traceability and verifiable compliance.

Representative Use Case. Existing implementations demonstrate how blockchain can support interoperability across multi-domain data spaces. Catena-X leverages the Gaia-X Digital Clearing House (GXDCH) to enable cross-space interoperability within the Gaia-X trust and compliance framework.²⁵

²⁴ <https://medium.com/@blockchainlabum/introducing-dsx-engine-truly-decentralized-dataspaces-connector-e40be91afe8c>

²⁵ <https://catenax-ev.github.io/docs/operating-model/how-data-space-operations>. Within this architecture, the Publication/Subscription service is transitioning from a traditional distributed messaging model toward a decentralized blockchain-based infrastructure.²⁶ Similarly, Ocean Protocol implements cross-chain bridges that enable token migration across multiple blockchain networks—including Ethereum, Polygon, Binance Smart Chain, and Fetch.ai—demonstrating how programmable interoperability mechanisms can facilitate cross-network asset exchange and data-service integration.²⁷ These examples highlight the emerging role of blockchain as a coordination and trust layer for federated data-space ecosystems.

²³ <https://docs.oceanprotocol.com/user-guides/basic-concepts/blockchain-the-backbone-of-ocean>

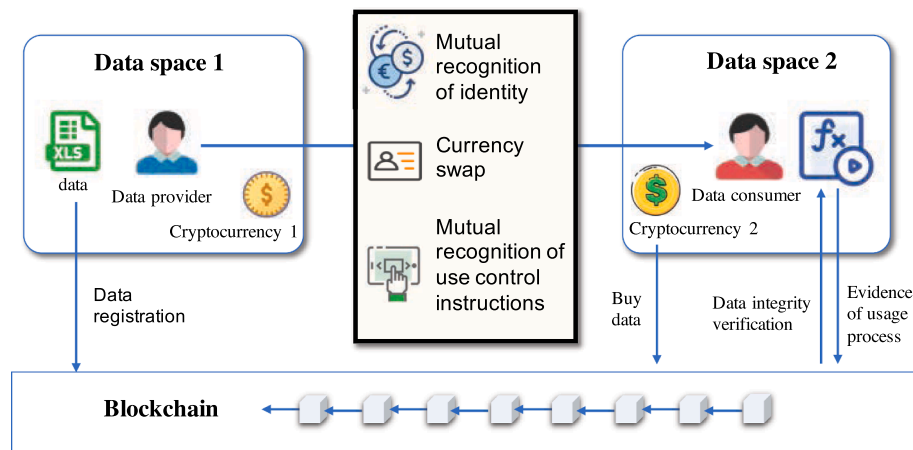


Fig. 12. Blockchain applications in crossing data spaces.

5. Challenges and future research directions of blockchain in supporting data spaces

This section identifies key open research challenges arising from the integration of blockchain with data spaces, aiming to inspire further exploration in this emerging field.

5.1. Interoperability challenges: Bridging heterogeneous governance and execution domains across data spaces

As data spaces evolve into federated ecosystems spanning multiple industries and jurisdictions, their underlying blockchain infrastructures increasingly exhibit deep heterogeneity not only at the protocol level but also at the governance and regulatory levels. Different data spaces may adopt public, consortium, or private blockchains with distinct identity management systems, compliance rules, and data usage semantics, resulting in fragmentation of trust and execution logic. Unlike conventional cross-chain interoperability problems, data spaces require *semantic interoperability*, where data usage policies, consent constraints, and liability definitions must remain consistent and enforceable across administrative domains. Existing approaches such as relay networks [87], interoperability protocols, and oracle systems primarily focus on asset or message transfer, but do not fully address cross-domain policy equivalence and verifiable governance alignment.

Future research should explore unified cross-space trust frameworks, policy-aware interoperability protocols, and verifiable identity and consent translation mechanisms that enable consistent enforcement of data usage rules across heterogeneous blockchain environments.

5.2. Privacy protection challenges: Reconciling immutable auditability with regulatory-compliant data sovereignty

Blockchain's transparency and immutability provide strong auditability guarantees, but they fundamentally conflict with the dynamic and jurisdiction-sensitive nature of data spaces, where data sovereignty and regulatory compliance (e.g., GDPR-style deletion rights and sector-specific confidentiality rules) are primary design constraints. The core tension lies in the fact that data spaces require *selective and revocable verifiability*, whereas blockchain enforces global and permanent replication. Although privacy-enhancing technologies such as zero-knowledge proofs, homomorphic encryption, secure multi-party computation, and trusted execution environments provide partial solutions, they often assume static computation models and do not naturally support evolving consent, policy updates, and long-lived data usage chains.

Future research should focus on cryptographic policy update mechanisms, verifiable revocation under immutable ledgers, and scalable

hybrid architectures that separate audit trails from sensitive semantic metadata while preserving end-to-end traceability.

5.3. Sustainability challenges: Balancing decentralized trust with lifecycle efficiency in large-scale data ecosystems

Blockchain introduces decentralized trust guarantees at the cost of persistent computational, storage, and synchronization overhead. In data spaces, this overhead is amplified by continuous metadata updates, fine-grained access logging, and long-lived data usage contracts across organizational boundaries. Unlike traditional blockchain applications, the workload is not event-driven but *continuously operational*, leading to sustained ledger growth and increasing validation complexity. Even energy-efficient consensus mechanisms such as PoS and BFT do not fully address the systemic cost of state accumulation and cross-domain replication.

Future research should move beyond consensus optimization toward *lifecycle-aware blockchain design*, including adaptive state pruning, verifiable off-chain archival systems, and incentive-compatible storage delegation mechanisms that jointly optimize trust, cost, and long-term system maintainability.

5.4. Data credential challenges: Representing and enforcing dynamic, multi-dimensional data usage rights

Although blockchain enables asset tokenization, data credentials in data spaces are fundamentally richer than traditional tokens, as they encode not only ownership but also fine-grained and context-dependent usage semantics. These include conditional access (e.g., purpose limitation), temporal constraints, transformation permissions, and cross-domain compliance obligations. Current token or NFT-based models are insufficient because they lack native support for *composable and stateful authorization*, where permissions evolve over time based on user behavior, policy updates, or regulatory changes. Moreover, data credentials must bridge on-chain representation and off-chain enforcement, ensuring that real-world data processing aligns with digitally defined constraints.

Future research should investigate programmable credential systems with verifiable policy execution, dynamic consent modeling, and integrated auditing mechanisms that jointly support identity, authorization, and accountability across the full data lifecycle.

5.5. Performance and scalability challenges: Enabling high-frequency, composable, and cross-domain data operations

Data spaces require continuous and fine-grained interactions, including metadata registration, contract negotiation, access verification, and

cross-domain orchestration. This leads to workload characteristics that differ significantly from conventional blockchain applications, particularly in terms of transaction frequency, state update velocity, and composability requirements. Existing scalability solutions such as Layer-2 networks, sharding, and off-chain computation primarily target transaction throughput but do not fully address the *composability latency* and *cross-domain state consistency* required in data spaces. In particular, ensuring consistent execution of chained data usage operations across heterogeneous domains remains an open problem.

Future research should explore modular blockchain architectures with separated execution and verification layers, asynchronous cross-domain state synchronization, and semantic-aware batching mechanisms to jointly optimize throughput, latency, and consistency guarantees.

5.6. AI integration challenges: Toward trustworthy and policy-aligned intelligence in data spaces

Integrating artificial intelligence into blockchain-enabled data spaces enables automated data discovery, intelligent negotiation, and adaptive governance, but also introduces new system-level tensions. Unlike centralized AI systems trained on static datasets, AI agents in data spaces must operate over distributed, heterogeneous, and policy-constrained data sources, where access rights, consent conditions, and usage constraints vary dynamically across domains. A key challenge lies in ensuring that AI-driven operations remain verifiable and compliant with fine-grained data usage policies. While blockchain can provide auditability for data access and model interactions, it does not inherently guarantee that AI inference, tool usage, or multi-step agent reasoning adheres to cross-domain constraints.

Future research should focus on blockchain-native AI governance mechanisms, including verifiable inference and execution protocols, policy-aware agent orchestration, and auditable AI pipelines that bind model behavior to enforceable data credentials. A promising direction is the co-design of data space infrastructures and AI models, enabling AI systems to operate directly over verifiable policies and consent structures, thereby achieving trustworthy and controllable intelligence in decentralized environments.

6. Conclusion

This paper presents a comprehensive survey of how blockchain technologies can support the construction and operation of data spaces. By reviewing existing academic research and industrial initiatives, the paper identifies eight key challenges faced by current data space systems, including identity management, data sovereignty, assetization, contract fulfillment, compliance, governance, security, and cross-space interoperability. Building on this analysis, seven fundamental capabilities of blockchain are synthesized and a layered architectural view of blockchain-enabled data spaces is organized to illustrate how blockchain mechanisms may support different functional layers such as identity management, data circulation, usage control, governance, and cross-space interaction. The survey further highlights representative technological approaches and emerging architectural patterns in recent studies and industrial practices. Finally, several open research challenges are outlined, including interoperability across heterogeneous data spaces, privacy-preserving data sharing, sustainable governance models, and system scalability. Overall, this survey provides a structured perspective and future research directions for building trustworthy and interoperable data spaces.

CRediT authorship contribution statement

Jianfeng Shi: Writing – review & editing, Writing – original draft, Conceptualization; **Lianggen Xu:** Investigation; **Feiwen Li:** Investigation; **Lingyuan Yin:** Writing – original draft; **Zihang Yin:** Writing –

review & editing; **Yang Liu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work is supported by the [National Key Research and Development Program of China \(No. 2023YFB2704200\)](#) and the [Guangxi Science and Technology Major Program \(No. GuiKeAA24263017\)](#).

Data availability

No data was used for the research described in the article.

References

- [1] R. Nagel, M.-K. Walda, J.P. Meyer, IntraDataspace: an architecture of a company-internal dataspace, in: *Proceedings of the 4th Eclipse Security, AI, Architecture and Modelling Conference on Data Space*, 2024, pp. 50–56.
- [2] S. Pretzsch, H. Drees, L. Rittershaus, Mobility data space: a secure data space for the sovereign and cross-platform utilization of mobility data, (2022) 343. Available at: https://link.springer.com/chapter/10.1007/978-3-030-93975-5_21.
- [3] D. Sarabia-Jácome, I. Lacalle, C.E. Palau, M. Esteve, Enabling industrial data space architecture for seaport scenario, in: *2019 IEEE 5th World Forum on Internet of Things (WF-IoT)*, IEEE, 2019, pp. 101–106.
- [4] A.S. Ahmadian, S. Franke, C. Gnoguem, J. Jürjens, Privacy-friendly sharing of health data using a reference architecture for health data spaces, in: *Proceedings of the 4th Eclipse Security, AI, Architecture and Modelling Conference on Data Space*, 2024, pp. 103–112.
- [5] G. Feng, X. Zhao, Y. Zhou, T. Xia, L. Jia, C. Xiong, Construction of a blockchain-based trusted data space for public health, in: *2025 International Conference on Algorithms, Data Mining, and Information Technology (ADMIT)*, IEEE, 2025, pp. 1–6.
- [6] J. Guo, Y. Cheng, D. Wang, F. Tao, S. Pickl, Industrial dataspace for smart manufacturing: connotation, key technologies, and framework, *Int. J. Prod. Res.* 61 (12) (2023) 3868–3883.
- [7] M. Neubauer, L. Steinle, C. Reiff, S. Ajdinović, L. Klingel, A. Lechler, A. Verl, Architecture for manufacturing-X: bringing asset administration shell, eclipse dataspace connector and OPC UA together, *Manuf. Lett.* 37 (2023) 1–6.
- [8] G. Farina, A. Kocian, G. Brunori, S. Chessa, M.B. Lai, D. Nardi, C. Schifanella, S. Bonura, N. Masi, S. Comella, et al., Interoperable traceability in agrifood supply chains: enhancing transport systems through IoT sensor data, blockchain, and dataspace, *Sensors* 25 (11) (2025) 3419.
- [9] Y. Song, W. Yang, S. Wei, Management in the energy data space: a blockchain and smart-contract-driven framework, *Int. J. Hous. Sci. Appl.* 47 (1) (2025).
- [10] G. DAL MAS, Data spaces: building trust in AI with blockchain.
- [11] T. Dam, L.D. Klausner, S. Neumaier, T. Priebe, A survey of dataspace connector implementations, (2023). [arXiv:2309.11282](https://arxiv.org/abs/2309.11282)
- [12] V. Siska, V. Karagiannis, M. Drobits, et al., Building a dataspace: technical overview, (2023). Available at: <https://publications.ait.ac.at/en/publications/building-a-dataspace-technical-overview-2/>.
- [13] M. Atzoria, A. Ciaramele, C. Diamantina, B. Di Martinoa, S. Distefano, T. Facchinetti, F. Montecchiana, A. Nocera, G. Ruffoa, R. Trasartia, Dataspace: concepts, architectures and initiatives, in: *CEUR Workshop Proceedings*, 3606, CEUR-WS, 2024.
- [14] M. Hauff, L.M. Comet, P. Moosmann, C. Lange, I. Chrysakis, J. Theissen-Lipp, FAIRness in dataspace: the role of semantics for data management, 3705 (2024). Available at: <https://biblio.ugent.be/publication/01J6SGPAD72CZQ0MWFF2BPKNXM>.
- [15] T. Schurig, A. Kari, D. Fürstenau, The symphony of orchestrated participatory data space governance: a systematic review, (2024). Available at: https://aisel.aisnet.org/ecis2024/track10_dmds_ecosystems/track10_dmds_ecosystems/9/?trk=public_post_comment-text.
- [16] S. Ishihara, T. Matsutsuka, Towards interoperable data spaces: comparative analysis of data space implementations between Japan and Europe, (2025). [arXiv:2501.15738](https://arxiv.org/abs/2501.15738)
- [17] S. Steinbuss, M. Punter, F. Fournier, I. Skarbovski, Blockchain technology in IDS, 2019.
- [18] W. Prinz, T. Rose, N. Urbach, Blockchain technology and international data spaces, 2022.
- [19] S.N. Khan, F. Loukil, C. Ghedira-Guegan, E. Benkhelifa, A. Bani-Hani, Blockchain smart contracts: applications, challenges, and future trends, *Peer-to-peer Netw. Appl.* 14 (2021) 2901–2925.
- [20] S. Yan, Analysis on blockchain consensus mechanism based on proof of work and proof of stake, in: *2022 International Conference on Data Analytics, Computing and Artificial Intelligence (ICDAI)*, IEEE, 2022, pp. 464–467.

- [21] F. Saleh, Blockchain without waste: proof-of-stake, *Rev. Financ. Stud.* 34 (3) (2021) 1156–1190.
- [22] X. Wang, S. Duan, J. Clavin, H. Zhang, Bft in blockchains: from protocols to use cases, *ACM Comput. Surv.* 54 (10s) (2022) 1–37.
- [23] S. Nakamoto, Bitcoin: a peer-to-peer electronic cash system (2008).
- [24] G. Wood, et al., Ethereum: a secure decentralised generalised transaction ledger, *Ethereum Proj. Yellow Pap.* 151 (2014) (2014) 1–32.
- [25] E. Androulaki, A. Barger, V. Bortnikov, C. Cachin, K. Christidis, A. De Caro, D. Enyeart, C. Ferris, G. Laventman, Y. Manevich, et al., Hyperledger fabric: a distributed operating system for permissioned blockchains, in: *Proceedings of the Thirtieth EuroSys Conference*, 2018, pp. 1–15.
- [26] H. Li, Y. Chen, X. Shi, X. Bai, N. Mo, W. Li, R. Guo, Z. Wang, Y. Sun, Fisco-bcos: an enterprise-grade permissioned blockchain system with high-performance, in: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2023, pp. 1–17.
- [27] X. Wei, S. Junyi, Z. Qi, C. Fu, Xuperchain: a blockchain system that supports smart contracts parallelization, in: *2020 IEEE International Conference on Smart Internet of Things (SmartIoT)*, IEEE, 2020, pp. 309–313.
- [28] H. Gong, Tokenised real-world assets (RWA) and decentralised physical infrastructure networks (dePIN), in: *The Web3 Revolution: Building the Future of Blockchain, DeFi, and the Metaverse*, Springer, 2024, pp. 175–197.
- [29] S.H. Alsamhi, A. Hawbani, A. Saif, E. Curry, Blockchain-driven decentralization for data sharing in the metaverse, in: *2025 4th International Conference on Computing and Information Technology (ICCIT)*, IEEE, 2025, pp. 610–616.
- [30] X. Sun, F.R. Yu, P. Zhang, Z. Sun, W. Xie, X. Peng, A survey on zero-knowledge proof in blockchain, *IEEE Netw.* 35 (4) (2021) 198–205.
- [31] S. Wu, J. Li, F. Duan, Y. Lu, X. Zhang, J. Gan, The survey on the development of secure multi-party computing in the blockchain, in: *2021 IEEE Sixth International Conference on Data Science in Cyberspace (DSC)*, IEEE, 2021, pp. 1–7.
- [32] P. Singh, M. Masud, M.S. Hossain, A. Kaur, Blockchain and homomorphic encryption-based privacy-preserving data aggregation model in smart grid, *Comput. Electr. Eng.* 93 (2021) 107209.
- [33] Y. Wang, J. Li, S. Zhao, F. Yu, Hybridchain: a novel architecture for confidentiality-preserving and performant permissioned blockchain using trusted execution environment, *IEEE Access* 8 (2020) 190652–190662.
- [34] S.H. Alsamhi, R. Myrzasheva, A. Hawbani, S. Kumar, S. Srivastava, L. Zhao, X. Wei, M. Guizan, E. Curry, Federated learning meets blockchain in decentralized data sharing: healthcare use case, *IEEE Internet Things J.* 11 (11) (2024) 19602–19615.
- [35] R. Chaganti, B. Bhushan, V. Ravi, A survey on blockchain solutions in DDoS attacks mitigation: techniques, open challenges and future directions, *Comput. Commun.* 197 (2023) 96–112.
- [36] L. Klug, W. Prinz, Fair prices for sustainability in agriculture and food. requirements and design options for a data-based transparency system, in: *Proceedings of the 24th Annual International Conference on Digital Government Research*, 2023, pp. 496–507.
- [37] N.B. García, M.H. Padilla, A.F.S. Gómez, Privacy preserving enablers for data space ecosystems, in: *International Summit on the Global Internet of Things and Edge Computing*, Springer, 2024, pp. 73–88.
- [38] R. Karl, V. Mazzonello, B. Pacheco, J. Silveira, S. Schlarb, V. Siska, A. Weissenfeld, Building a data space for the public security domain using smart contracts, in: *2024 4th Intelligent Cybersecurity Conference (ICSC)*, IEEE, 2024, pp. 32–38.
- [39] A.D. Leander, Identification and Implementation of Suitable Decentralized Identifier Methods for Self-Sovereign Identity Wallets in a Data Space, Master's thesis, University of Twente, 2024.
- [40] S. Gabrielli, S. Krenn, D. Pellegrino, J.C. Pérez Bañ, P. Pérez Berganza, S. Ramacher, W. Vandevelde, Kraken: a secure, trusted, regulatory-compliant, and privacy-preserving data sharing platform, in: *Data Spaces: Design, Deployment and Future Directions*, Springer International Publishing Cham, 2022, pp. 107–130.
- [41] D.S.B. Alliance, Technical convergence discussion document, 2023.
- [42] H. Yan, B. Chen, Architecture of integrated resource system based on dataspace, in: *International Conference on Computer Engineering and Networks*, Springer, 2023, pp. 74–85.
- [43] AMable, Guide to blockchain for additive manufacturing, 2019, AMable project consortium.
- [44] Ocean Protocol Foundation Ltd., Ocean Docs. <https://docs.oceanprotocol.com/>.
- [45] S. Meneguzzo, A. Favenza, V. Gatteschi, C. Schifanella, Exploring the potential of energy data marketplaces: an approach based on the ocean protocol, in: *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, IEEE, 2023, pp. 1488–1494.
- [46] S. Meneguzzo, S. Bergia, A. Favenza, C. Schifanella, A. Mozzato, V. Gatteschi, Shaping the future of energy markets: federated systems and blockchain synergy, in: *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, IEEE, 2024, pp. 2026–2031.
- [47] S. Meneguzzo, A. Favenza, V. Gatteschi, C. Schifanella, Integrating a dlt-based data marketplace with idsa for a unified energy dataspace: towards silo-free energy data exchange within gaia-x, in: *2023 5th Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS)*, IEEE, 2023, pp. 1–2.
- [48] K. Talmoudi, K. Choukri, I. Gavanon, Compliance by design methodologies in the legal governance schemes of european data spaces, in: *Proceedings of the Workshop on Legal and Ethical Issues in Human Language Technologies@ LREC-COLING 2024*, 2024, pp. 1–5.
- [49] O. Gallay, K. Korpela, N. Tapio, J.K. Nurminen, A peer-to-peer platform for decentralized logistics, in: *Proceedings of the Hamburg International Conference of Logistics (HICL)*, epubli, 2017, pp. 19–34.
- [50] M.A. Diallo, Z. Li, J. Sun, X. Zhang, D. Fu, Optimizing logistics operations: an international data space (IDS) inspired multi-sided data space marketplace for enhanced data sovereignty in the logistics industry, in: *Proceedings of the 3rd International Conference on Big Data Economy and Digital Management, BDEDM 2024*, January 12–14, 2024, Ningbo, China, EAI, 2024.
- [51] K. Kyriakou, A. Apostolaras, P. Velentzas, G. Benos, K. Koutsoukos, C. Symvoulidis, K. Liang, Z. Shi, A. Leonidis, K. Miniadou, et al., A secure and trustworthy biometric data ecosystem for cross-border suspect identification, in: *2024 IEEE International Conference on Big Data (BigData)*, IEEE, 2024, pp. 2762–2771.
- [52] K. Kyriakou, A. Apostolaras, P. Velentzas, I. Syrigos, S. Maglaveras, S. Evangelatos, E. Veroni, T. Korakis, Biometrics data space: ensuring trustworthy and secure data exchange for suspect identification, in: *2024 5th International Conference in Electronic Engineering, Information Technology & Education (EETEE)*, IEEE, 2024, pp. 1–6.
- [53] S.H. Alsamhi, E. Curry, A. Hawbani, S. Kumar, U.U. Hassan, N.S. Rajput, DataSpace in the sky: a novel decentralized framework to secure drones data sharing in B5G for industry 4.0 toward industry 5.0, *Ind. Manuf. Eng.* (2023).
- [54] S.H. Alsamhi, S. Srivastava, M. Rashid, A. Alhabeed, S. Kumar, N.S. Rajput, A. Hawbani, L. Zhao, M.A.A. Al-qaness, E. Curry, Skyward secure: advancing drone data-sharing in 6G with decentralized dataspace and supported technologies, *J. Parallel Distrib. Comput.* 199 (2025) 105040.
- [55] L. Dutkiewicz, Y. Miadzwetskaya, H. Ofé, A. Barnett, L. Helminger, S. Lindstaedt, A. Trügler, Privacy-preserving techniques for trustworthy data sharing: opportunities and challenges for future research, (2022) 319–335. Available at: https://link.springer.com/chapter/10.1007/978-3-030-98636-0_15.
- [56] G.M. Sang, L. Xu, P. De Vrieze, Y. Bai, F. Pan, Predictive maintenance in industry 4.0, in: *Proceedings of the 10th International Conference on Information Systems and Technologies*, 2020, pp. 1–11.
- [57] M. Gall, TrackChain: blockchain-basiertes track & trace in supply chains mit dem industrial data space (2019).
- [58] H. Liu, W. Tai, Y. Wang, S. Wang, A blockchain-based spatial data trading framework, *EURASIP J. Wirel. Commun. Netw.* 2022 (1) (2022) 71.
- [59] G. Farina, G. Brunori, S. Chessa, A. Kocian, M.B. Lai, D. Nardi, C. Schifanella, S. Bonura, N. Masi, F. Ambrosino, et al., Interoperable traceability in supply chains: a use case in agritech, in: *International Conference on Intelligent Transport Systems*, Springer, 2024, pp. 29–45.
- [60] A. Kovach, J. Lanza, L. Montalvillo, A. Urbieta, Sovereign IIoT data exchange using DAG-Based DLT and international data spaces architecture, in: *Proceedings of the 4th Eclipse Security, AI, Architecture and Modelling Conference on Data Space*, 2024, pp. 76–85.
- [61] A. Kovach, L. Montalvillo, A. Urbieta, J. Lanza, IOTA-enabled decentralized data space for IIoT ecosystems (2024).
- [62] Y.R. Witharanage, S. Figueroa-Lorenzo, S. Arrizabalaga, An HFB interface for the adoption of blockchain in data spaces, *Comput. Sci. Inf. Technol. ISSN 14* (2024) 2231–5403. Available at: <https://www.csitp.net/abstract/14/1422csit06>.
- [63] P. Singh, M.J. Beliatas, M. Presser, Enabling edge-driven dataspace integration through convergence of distributed technologies, *Internet Things* 25 (2024) 101087.
- [64] S.H. Alsamhi, A. Hawbani, S. Kumar, M. Timilsina, M. Al-Qatf, R. Haque, F. Nashwan, L. Zhao, E. Curry, Empowering dataspace 4.0: unveiling promise of decentralized data-sharing, *IEEE Access* 12 (2024) 112637–112658. Available at: <https://ieeexplore.ieee.org/abstract/document/10634457>.
- [65] Eclipse Contributors, EDC documentation. <https://eclipse-edc.github.io/>.
- [66] DeltaDAO, Pontus-X documentation. <https://docs.pontus-x.eu/>.
- [67] International Data Spaces Association, IDS-RAM 4. <https://docs.internationaldataspaces.org/ids-knowledgebase/ids-ram-4>.
- [68] S. Dalmolen, H. Bastiaansen, H. Moonen, W. Hofman, M. Punter, E. Cornelisse, Trust in a multi-tenant, logistics, data sharing infrastructure: opportunities for blockchain technology, in: *5th International Physical Internet Conference, IPIC 2018*, University of Groningen, 2018, pp. 299–309.
- [69] V. Barak, I. Paraskakis, S. Veloudis, P. Yadav, Responsible information sharing in the era of big data analytics facilitating digital economy through the use of blockchain technology and observing GDPR, in: *CLOSER*, 2024, pp. 257–264.
- [70] Catena-X, Catena-X doc. <https://catena-x.net/>.
- [71] A.R. Costagliola, C. Mazzocca, A. Bujari, R. Montanari, P. Bellavista, VESPACE: a verifiable blockchain-based data space solution to empower the data economy, *Comput. Commun.* 239 (2025) 108180.
- [72] H. Seike, Y. Aoki, N. Koshizuka, Quantitative analysis of blockchain consensus effects on dataspace transaction latency, in: *2025 IEEE International Conference on Big Data (BigData)*, IEEE, 2025, pp. 3862–3871.
- [73] F. Gentili, Blockchain Integration in Data Space Architectures: the NOUS Project, Ph.D. thesis, Politecnico di Torino, 2025.
- [74] Z.-Y. Liang, G.-Y. Liu, Y. Ren, M. Yang, R.-W. Jiang, Y. Luo, Y.-S. Ma, Trustworthy data space collaborative trust mechanism driven by blockchain: technology integration, cross-border governance, and standardization path, *Information* 16 (12) (2025) 1066.
- [75] D.V. Villaverde, C.Á. López, Enhancing environmental sustainability: decentralized dataspace architecture via DAO and oracles, in: *Distributed Computing and Artificial Intelligence, Special Sessions I, 21st International Conference, Springer Nature*, 2025, p. 409.
- [76] J. Cano-Benito, A. Cimmino, S. Hertling, H. Paulheim, R. García-Castro, Analysing semantic data storage in Distributed Ledger Technologies for Data Spaces, (2025). [arXiv:2507.07116](https://arxiv.org/abs/2507.07116)
- [77] X. Yang, Q. Luo, J. Xu, Q. Cao, Data secure storage mechanism for trustworthy data space, *Electronics* 14 (21) (2025) 4348.
- [78] J. Shi, Y. Zhang, L. Yin, C. Chi, Y. Liu, DSRep: implementing decentralized reputation evaluation for data space based on a low-cost and high-throughput blockchain, *Results Eng.* 27 (2025) 105672. Available at: <https://www.sciencedirect.com/science/article/pii/S2590123025017438>.

- [79] Y. Chen, C. Liu, Y. Wang, Y. Wang, A self-sovereign decentralized identity platform based on blockchain, in: 2021 IEEE Symposium on Computers and Communications (ISCC), IEEE, 2021, pp. 1–7.
- [80] J. Sedlmeir, R. Smethurst, A. Rieger, G. Fridgen, Digital identities and verifiable credentials, *Bus. Inf. Syst. Eng.* 63 (5) (2021) 603–613.
- [81] H. Taherdoost, Non-fungible tokens (NFT): a systematic review, *Information* 14 (1) (2022) 26.
- [82] S.K. Ezzat, Y.N.M. Saleh, A.A. Abdel-Hamid, Blockchain oracles: State-of-the-art and research directions, *IEEE Access* 10 (2022) 67551–67572.
- [83] R. Shi, R. Cheng, B. Han, Y. Cheng, S. Chen, A closer look into IPFS: accessibility, content, and performance, *Proc. ACM Meas. Anal. Comput. Syst.* 8 (2) (2024) 1–31.
- [84] S. Williams, V. Diordiiev, L. Berman, I. Uemlianin, Arweave: a protocol for economically sustainable information permanence, (2019). Available at: <https://arweave.org/yellow-paper.pdf>.
- [85] L.T. Khrais, Comparison study of blockchain technology and IOTA technology, in: 2020 Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC), IEEE, 2020, pp. 42–47.
- [86] K. Saurabh, P. Upadhyay, N. Rani, Towards blockchain decentralized autonomous organizations (DAO) design, *Inf. Syst. Front.* (2024) 1–23.
- [87] P. Frauenthaler, M. Sigwart, C. Spanring, M. Sober, S. Schulte, ETH relay: a cost-efficient relay for ethereum-based blockchains, in: 2020 IEEE International Conference on Blockchain (Blockchain), IEEE, 2020, pp. 204–213.